

United International University
Department of Computer Science and Engineering

CSE-3313: Computer Architecture

Final Examination: Spring 2026

Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from
the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. (a) Modify and annotate the block diagram for the single-cycle data path and **write the control unit values** for the following R type instruction: '**cs \$s0, \$s1, \$s2**'. The task of the instruction is to find out which value (stored at the registers) is higher between \$s1 and \$s2. Afterwards, the identified greater value will be multiplied by 8 and stored at \$s0. The value of \$s0, \$s1, \$s2 are stored in register rd, rs, and rt in the machine code translation. The machine code format for the instruction is given below in Table 1: [5]

opcode	rs	rt	rd	shamt	function code
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits

Table 1: Machine code format of instruction '**cs \$s0, \$s1, \$s2**'.

- (b) Modify and annotate the block diagram for the single-cycle data path and **write the control unit values** for this instruction: '**ds \$s0, \$s1, 255**'. The task of this instruction is to divide the offset value, which is 255 here by 64. The division result should then be subtracted from the value stored at \$s1. The subtraction result will give an address. Read the value from that address and store it in \$s0. The value of the register number of \$s0 and \$s1 is stored at rt and rs, and the constant value 255 is stored in the offset. The machine code format for ds is given in Table 2: [5]

op	rs	rt	offset
6 bits	5 bits	5 bits	16 bits

Table 2: Machine code format of instruction '**ds \$s0, \$s1, 255**'.

2. Consider a processor that goes through the following 5 stages in Table 3, while executing an instruction.

Pipeline Stage	Delay
Instruction Fetch (IF)	100 ns
Instruction Decode (ID)	120 ns
Execute (EX)	200 ns
Memory Access (MEM)	180 ns
Write Back (WB)	150 ns

Table 3: Five-stage Pipeline.

- (a) Observe the following code. Assume that the base address of array A = {5, 8, 16} is stored at \$s1. For the following code, determine the time required for the execution of this code using a single cycle processor. [3]

```
1      addi $s0, $zero, 2
2      addi $s1, $zero, 1000
3      LOOP:
4          sll $s0, $s0, 1
5          lw  $t0, 0($s1)
6          addi $s1, $s1, 4
7          bne $s0, $t0, LOOP
```

Code for 2(a)

```

1      lw    $t0, 0($s0)
2      sw    $t1, 4($s0)
3      add   $t2, $t0, $t1
4      sll   $t3, $t1, $s2
5      andi  $t2, $t3, $t9

```

Code for 2(b), 2(c)

- (b) Determine the execution time of the above code using the basic pipelining method, and illustrate your answer with a timing diagram. [4]
- (c) Determine the optimized execution time and illustrate it with a timing diagram. [4]
- (d) Can control hazards be eliminated using code scheduling? If so, explain with an example. [4]
3. A CPU uses a direct-mapped cache and generates 20-bit physical addresses. The main memory is byte-addressable. The cache has a capacity of 4 KB, and each cache block contains 4 words.

Index	V	D	Tag	Data
0	0	0	–	Empty
...	...	...	...	...
128	1	0	2	9216–9223
136	1	0	2	9280–9287
140	1	1	1	5216–5223
141	1	0	0	1128–1135
...	...	...	...	...
$2^n - 1$	0	0	–	Empty

Table 4: Initial Cache Entries.

- (a) Calculate the actual size of the cache in bits. [4]
- (b) Find the miss ratio if the following memory addresses are accessed in this order Assume the cache initially contains only the entries shown in the Table 4. All other cache lines are empty. [8]

9284	5220	9300	11200	5232	13380	13377
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- (c) The CPU accesses byte address 154372. Write the corresponding 20-bit physical address (memory field) and divide it into the Tag, Index, and Block Offset fields [3]